Exercises for Chapter 4

Multiple Choice Questions

1. **Which of the following statements about the normal distribution is *false*?**
   (a) The normal distribution is symmetric about its mean.
   (b) Approximately 95% of the data in a normal distribution lie within one standard deviation of the mean.
   (c) The total area under the normal distribution curve is equal to 1.
   (d) The normal distribution is completely determined by its mean and standard deviation.

2. **If a dataset is normally distributed with mean μ and standard deviation σ, the Z-score of an observation x is calculated as:**
   (a) $Z = \frac{x - \mu}{\sigma}$
   (b) $Z = \frac{\mu - x}{\sigma}$
   (c) $Z = (x - \mu) \times \sigma$
   (d) $Z = (\mu - x) \times \sigma$

3. **The standard normal distribution is a normal distribution with:**
   (a) Mean of 0 and standard deviation of 1.
   (b) Mean of 1 and standard deviation of 1.
   (c) Mean of 0 and variance of 1.
   (d) Mean of 1 and variance of 0.

4. **The empirical rule states that for a normal distribution, approximately what percentage of data falls within two standard deviations of the mean?**
   (a) 68%
   (b) 95%
   (c) 99.7%
   (d) 100%

5. **Which of the following is *not* a property of a binomial distribution?**
   (a) Fixed number of trials.
   (b) Each trial has only two possible outcomes.
   (c) Trials are independent.
   (d) Probability of success varies from trial to trial.

6. **In a geometric distribution, the random variable X represents:**
   (a) The number of successes in a fixed number of trials.

(b) The number of failures until the first success.
(c) The probability of success in each trial.
(d) The total number of trials.

7. **The mean and variance of a binomial distribution with parameters n and p are:**

   (a) Mean $= np$, Variance $= np(1-p)$
   (b) Mean $= np$, Variance $= n^2p(1-p)$
   (c) Mean $= np$, Variance $= np^2(1-p)$
   (d) Mean $= n(1-p)$, Variance $= np(1-p)$

8. **Which of the following statements is true about the normal approximation to the binomial distribution?**

   (a) It can be used when $np \geq 10$ and $n(1-p) \geq 10$.
   (b) It can be used for any binomial distribution regardless of n and p.
   (c) It cannot be used when p is close to 0.5.
   (d) It is exact and does not require a continuity correction.

9. **The negative binomial distribution models:**

   (a) The number of trials until the first success.
   (b) The number of successes in a fixed number of trials.
   (c) The number of trials until a fixed number of successes occur.
   (d) The probability of k successes in n trials.

10. **In a right-skewed distribution, most Z-scores are:**

    (a) Positive
    (b) Negative
    (c) Zero
    (d) Undefined

Exercises for Chapter 4: Distributions of Random Variables

1. The heights of adult men in a certain population are approximately normally distributed with a mean of 70 inches and a standard deviation of 3 inches. **What proportion of men are taller than 73 inches?**

2. Test scores on a college entrance exam are normally distributed with a mean of 500 and a standard deviation of 100. **What is the minimum score a student must achieve to be in the top 10% of test-takers?**

3. A student scores 85 on a test where the class mean is 75 with a standard deviation of 5. **What is the student's Z-score?**

4. Marathon finishing times are normally distributed with a mean of 240 minutes and a standard deviation of 30 minutes. **What percentile is a runner in if they finish in 195 minutes?**

5. For a normally distributed variable with mean 100 and standard deviation 15, **what value corresponds to the 75th percentile?**

6. IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. **What IQ score corresponds to the top 2% of the population?**

7. The weights of apples in an orchard are normally distributed with a mean of 150 grams and a standard deviation of 10 grams. **Approximately what percentage of apples weigh between 140 grams and 160 grams?**

8. Exam scores are normally distributed with a mean of 78 and a standard deviation of 6. **What percentage of students scored between 72 and 90?**

9. The lengths of nails produced by a machine are normally distributed with a mean of 5 cm and a standard deviation of 0.1 cm. **What proportion of nails are shorter than 4.8 cm?**

10. The lifespan of a certain brand of light bulbs is normally distributed with a mean of 1,200 hours and a standard deviation of 100 hours. **What is the probability that a randomly selected bulb lasts between 1,050 and 1,350 hours?**

11. A fair die is rolled repeatedly until a six appears. **What is the probability that the first six appears on the third roll?**

12. A website has a 20% chance of crashing each time it is accessed. **What is the expected number of accesses before the site crashes?**

13. A student randomly guesses on each of 10 multiple-choice questions, each with 4 options. **What is the probability that they get exactly 3 questions correct?**

14. A batch of 1,000 items contains 2% defective items. **If a sample of 50 items is randomly selected, what is the probability that the sample contains at most one defective item?**

15. A fair coin is tossed 100 times. **What is the approximate probability that it lands heads more than 60 times?**

16. In a city, 30% of households own a dog. **If 200 households are randomly surveyed, what is the probability that at least 75 households own a dog?**